Stephen Hung

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Education

University of California, Berkeley GPA: 3.66/4.00 Expected: May 2028

Bachelor of Science in Electrical Engineering and Computer Sciences

- **Coursework:** Data Structures, Algorithms, Computer Architecture, Discrete Math, Computer Security, Artificial Intelligence
- **Awards/Activities:** Dean's List, Cal Theta Tau (Professional Engineering Fraternity), Cal Blueprint (Nonprofit Tech Club)

Skills

Languages: Python, JavaScript (ES6+), TypeScript, Java, C++, C, Ruby, SQL, Rust, HTML/CSS, Bash, Assembly, MATLAB
AI/ML: PyTorch, NumPy, Pandas, LangChain, OpenAI/Claude/Gemini APIs, LLMs, RAG, Computer Vision, YOLOv8, SAM
Frameworks & Tools: React.js/Native, Node.js, FastAPI, Next.js, Tailwind CSS, Docker, Git GitHub Actions, REST APIs
Databases & Cloud: PostgreSQL, MongoDB, Supabase, AWS (S3, EC2, Lambda), GCP (Cloud Storage, Vertex AI), SQLite

Experience

AI/ML Engineer Intern, Walmart Global Tech | *Sunnyvale, CA* Jun 2026 - Present

- Built **GTAP (Global Testing Automation Platform)**, a production AI testing platform for **200+ Walmart engineering teams** indexing thousands of repositories with **96.5% Recall@8 and 0.913 MRR** using hybrid FTS5 and embedding retrieval.
- Reduced peak ingestion memory **1,000×** and increased embedding throughput **8×** by replacing batch loading with streaming ingestion and parallel embedding batches; eliminated **80%** of redundant database writes through content-hash deduplication.
- Shipped **35 MCP tools and 27 REST endpoints** with multi-tenant authentication and production Kubernetes deployment.

Software Developer, Cal Blueprint (Replate Food Rescue) | *Berkeley, CA* Sept 2025 – Present

- Building **React Native** and **Expo** app that replaces Replate's pen-and-paper donation logging workflow, eliminating manual re-entry that delays metrics reporting across a network rescuing **1M+ lbs** of surplus food annually for **80+** recipient nonprofits.

Software Engineering Consultant, ClearPath Medical | *Tustin, CA (Remote)* Sept 2025 – Dec 2025

- Developed **AI powered PFMEA automation tool** using **FastAPI** and **LLMs** with **self validating agentic pipeline** that converts work instruction PDFs into risk assessments, reducing analysis time for FDA compliant medical device QC by **85%**.

Software Engineering Intern, OptiGenix | *Berkeley, CA* May 2025 – Aug 2025

- Achieved **92.3% extraction accuracy** on labeled test data by training a **generative AI** model using Google Vertex AI, structuring detailed blood marker data from **60 unstructured** blood test PDFs stored and processed in **GCP cloud storage**.
- Designed and deployed **HIPAA compliant data workflows** on **GCP**, supporting **7,500+** monthly PDF uploads with Cloud Storage, **IAM role based access control**, server side encryption, and audit logging for regulatory compliance and traceability.

Projects

yolodex (Agentic YOLO Training) - OpenAI Codex Hackathon Winner | *YOLOv8, Codex* *Live Demo*

- Built Codex-native agentic pipeline that transforms raw unlabeled video into trained **YOLOv8** object detection models, automating frame extraction, labeling, augmentation, training, and evaluation with configurable **mAP@50** accuracy targets.
- Engineered **parallel subagent orchestration** spawning configurable Codex subagents in isolated git worktrees for distributed data labeling, with an autonomous iterative train-eval loop that re-labels and retrains until target accuracy threshold is met.
- Presented on stage at **OpenAI HQ** to judges including **Sam Altman**, **Greg Brockman**, **Sonya Huang** (Sequoia), and **Lenny Rachitsky**; featured on OpenAI's official socials, generating **500K+ impressions** and **20K+ reactions** in 24 hours.

opal (AI Gaming Companion) - $250K Grand Prize | *LiveKit, YOLOv8, Claude CUA* *Devpost* | *Live Demo*

- Won **$250,000 grand prize** pitching live to **Tim Draper**. Built autonomous **AI gaming agent** with game agnostic architecture for FPS and strategy games, joining voice channels and playing as a real teammate, growing **1000+ members**.
- Trained custom **YOLOv8** and depth estimation model at **60+ FPS** for enemy detection, **Claude** for strategic decision making, **LiveKit WebRTC** for real time voice at **89ms latency**, with cloud gaming infrastructure to ensure full anti cheat compliance.

vima (Spatial Intelligence for VLMs) - 2x Hacktech 2026 Winner | *FastAPI, Gemini, SAM* *Devpost* | *Live Demo*

- Won **Ironsite Prize** for construction intelligence platform turning footage into auditable spatial memory via **FastAPI** pipeline (SAM masks, depth estimation, COLMAP-style spatial zones), achieving **+33.2%** over VLM-only baseline on spatial reasoning.
- Engineered **object-event episodic memory** producing 118 temporal episodes with cited frame retrieval, exposed via **Next.js** dashboard, **MCP server**, and CLI; built mobile field companion with Solana reward layer for swipe-based claim verification.

flow (Spatial Learning Platform) - SB Hacks XII 2x Winner | *ElevenLabs, Marble, Gemini* *Devpost* | *Live Demo*

- Won **President's Pick and MLH Best Use of ElevenLabs** for spatial learning platform where users speak concepts and explore photorealistic 3D Gaussian Splat environments with AI narrated overlays and contextual Q&A powered by **Gemini**.
- Orchestrated **6 API async pipeline** with **WebSocket progress tracking** through end-to-end 5 minute generation (Deepgram, Gemini, Marble, SparkJS), scene caching, rate limiting, and persistent user world library on **Vultr/MongoDB Atlas**.

iris (Natural-Language Video Editor) - 2nd Place CitrusHack 2026 | *Gemini, Veo, SAM2* *Devpost* | *Live Demo*

- Built natural-language video editor where **Gemini 2.5 Flash** orchestrates **SAM2** segmentation and **Veo 3.1** generation; agent identifies regions, scores on visual coherence + temporal consistency, then propagates edits to every appearance of the subject.
- Designed **20-tool Gemini function-calling agent** streamed over SSE (preview, split, grade, remix, batch) with **PostgreSQL** conversation persistence; shipped a CLI so external AI agents (Claude Code, Codex) can drive the editing pipeline autonomously.

darwin (Multi Agent Coding) - Cal Hacks 12.0 Winner | *Sui, Letta, LiveKit* *Devpost* | *Live Demo*

- Built **6 agent orchestration system** where 4 AI coding agents generate UIs from natural language while a commentator agent analyzes outputs, with **Letta cross agent memory** for context sharing and live narration across the agent swarm.